Springboard Virtual Internship 6.0

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Branch: Computer Science and Engineering

Domain: Data Visualization

Project: Visualizing U.S. Natural Disaster Declarations

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Milestone-04: Incident Type Analysis

1. Objective

The objective of this milestone is to perform a detailed analysis of disaster incident types and understand their distribution across different states and assistance programs using the FEMA disaster declarations dataset.

This milestone focuses on identifying the most frequent disaster types, analyzing how disaster types vary across states, and understanding how different disasters are associated with assistance programs.

Additionally, this stage integrates insights from temporal and geographic analysis to provide a comprehensive understanding of disaster patterns.

Through this analysis, we aim to identify dominant disaster types, regional variations, and the relationship between disaster types and assistance programs.

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2. Problem Formulation

Analysis Type

This milestone involves the following types of analysis:

• Exploratory Data Analysis (EDA)
• Comparative Analysis
• Categorical Data Analysis

The goal is to analyze disaster incident types and extract meaningful insights using visualizations.

Key Questions

The following questions guide the incident type analysis:

• Which disaster types occur most frequently?
• Which states experience specific disaster types the most?
• How do disaster assistance programs relate to different incident types?
• What key insights can be derived from the overall analysis?

These questions help in understanding disaster composition and impact.

3. Dataset Preparation for Incident Analysis

Input Dataset

The dataset used for this analysis is:
DatasetDeclarationsSummaries.csv

This dataset was cleaned and processed in previous milestones.

Required Columns Used

The following columns were used for the analysis:

- incidentType – Represents the type of disaster
- state – Represents the state where the disaster occurred
- year – Year of the disaster declaration
- month – Month of the disaster declaration
- ihProgramDeclared – Indicates Individual Housing assistance
- paProgramDeclared – Indicates Public Assistance programs

Additional optional columns include:

- declarationDate
- stateName

Data Validation

Before performing analysis, the dataset was validated to ensure consistency.

The following checks were performed:

- Verified that incident types were consistent
- Checked for missing values in key columns
- Ensured records exist across multiple states and disaster types

These steps ensured that the dataset was suitable for incident type analysis.

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4. Incident Type Aggregation

Data aggregation was performed to analyze disaster types and their distribution.

4.1 Incident Type Aggregation

The dataset was grouped by incident type to calculate the total number of disaster declarations for each type.

This aggregation helps identify:

• Most frequent disaster types
• Overall disaster composition

4.2 State + Incident Type Aggregation

The dataset was grouped by:

• State
• Incident Type

This aggregation helps determine:

• Which disaster types dominate in specific states
• Regional distribution of disaster types

4.3 Incident Type vs Assistance Aggregation

The dataset was grouped by incident type and assistance indicators:

• ihProgramDeclared
• paProgramDeclared

This helps analyze how disaster types are related to assistance programs.

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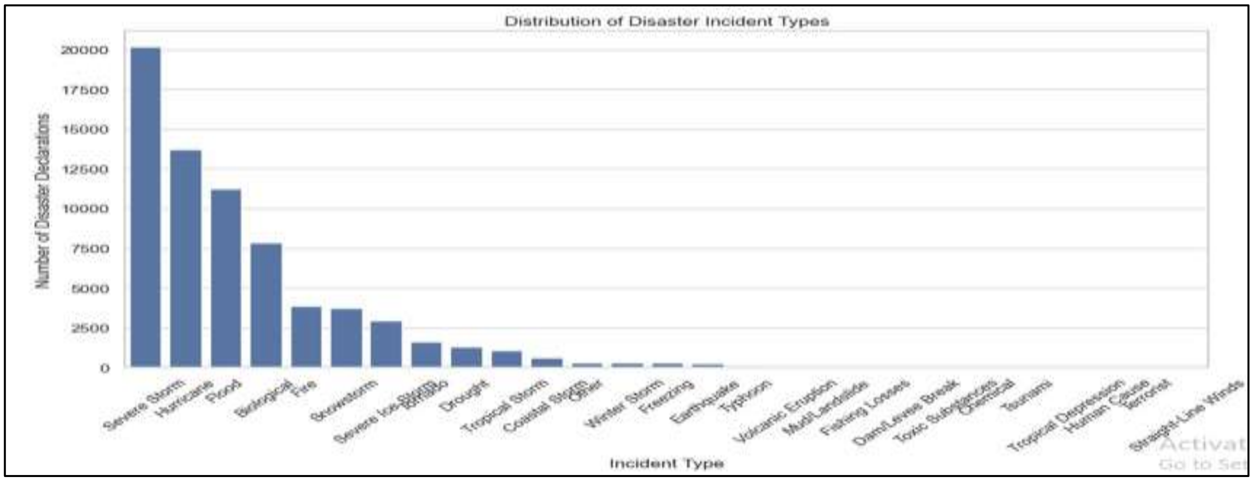
5. Visualizations

Several visualizations were created to analyze disaster incident types.

5.1 Distribution of Disaster Incident Types

This bar chart displays the total number of disaster declarations for each incident type by aggregating the dataset based on the incidentType column.

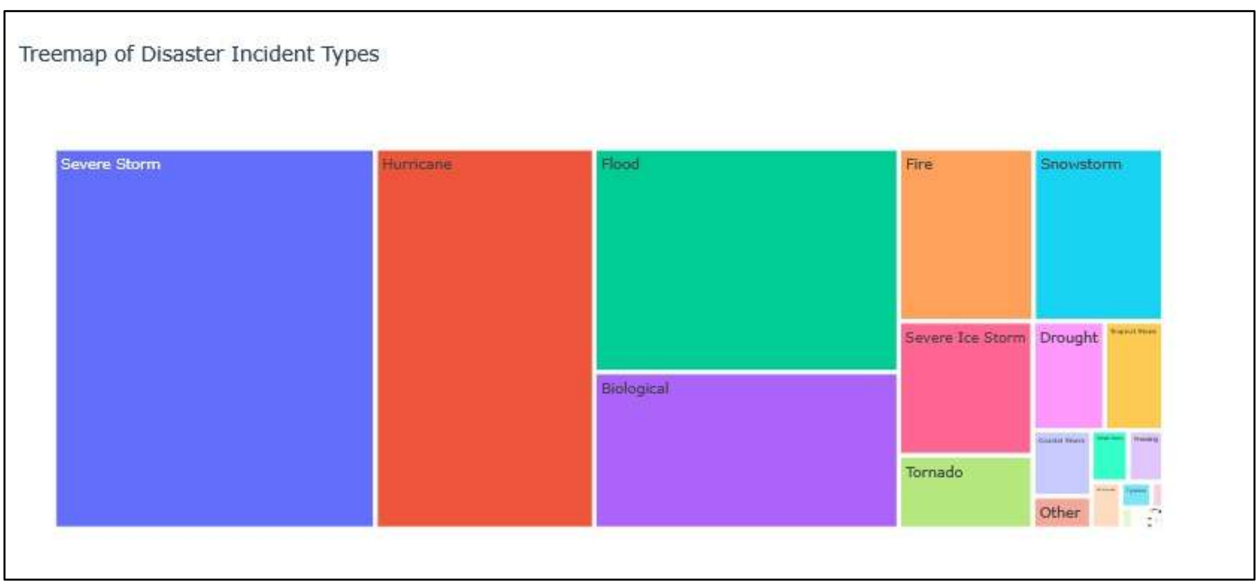


Interpretation

This bar chart represents the frequency of each disaster type.
It shows that certain disaster types occur more frequently than others.
Weather-related disasters dominate the dataset, indicating their high occurrence rate.

5.2 Treemap of Disaster Incident Types

This treemap visualizes the proportion of different disaster types using hierarchical blocks, where each block size represents the frequency of that incident type.

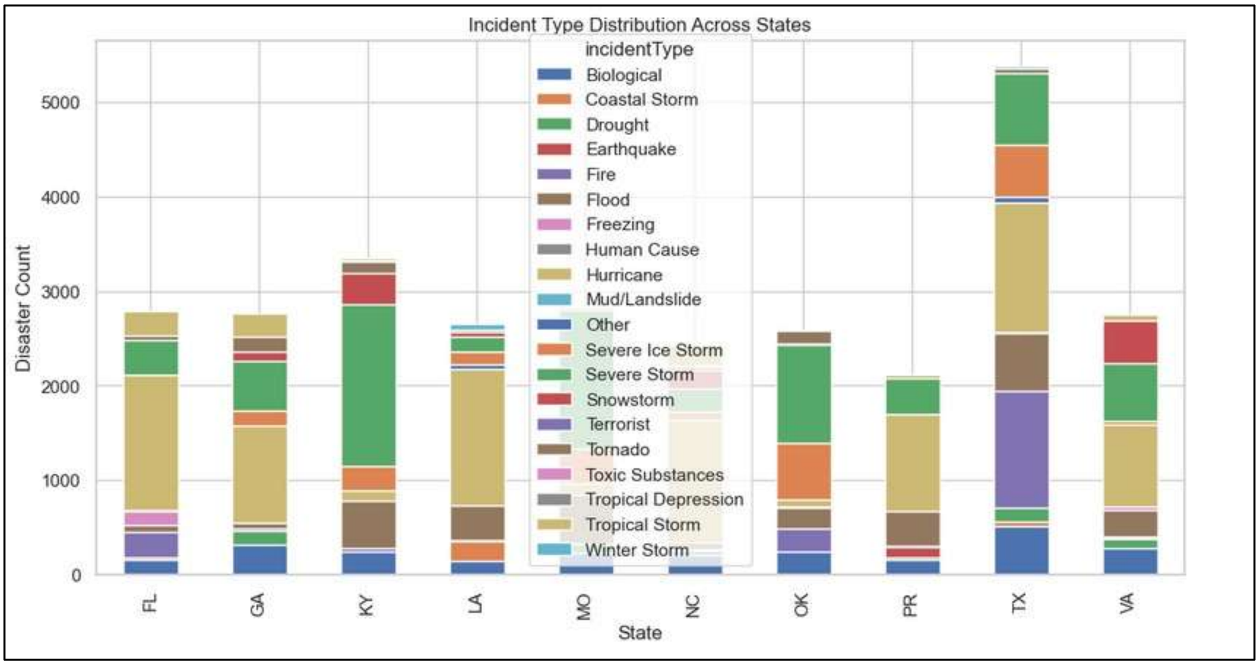


Interpretation

The treemap visualizes disaster types based on their frequency using hierarchical blocks.
Larger blocks indicate more frequent disaster types.
This visualization provides a quick and effective comparison of disaster composition.

5.3 State vs Incident Type Distribution

This stacked bar chart represents the distribution of different disaster types across states by grouping the dataset using state and incidentType.

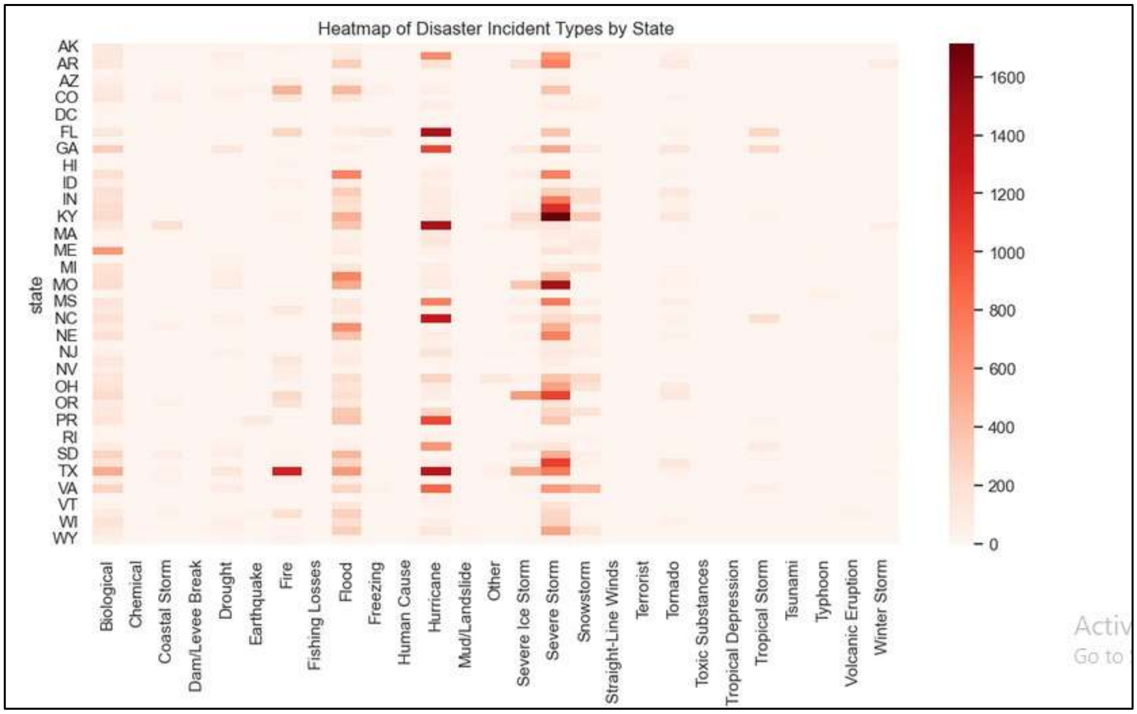


Interpretation

This stacked bar chart shows how different disaster types are distributed across states.
Different states experience different types of disasters.
It helps identify regional patterns and dominant disaster types in specific states.

5.4 Heatmap of Disaster Incident Types by State

This heatmap shows the frequency of disaster types across states using a matrix format, where rows represent states and columns represent incident types.

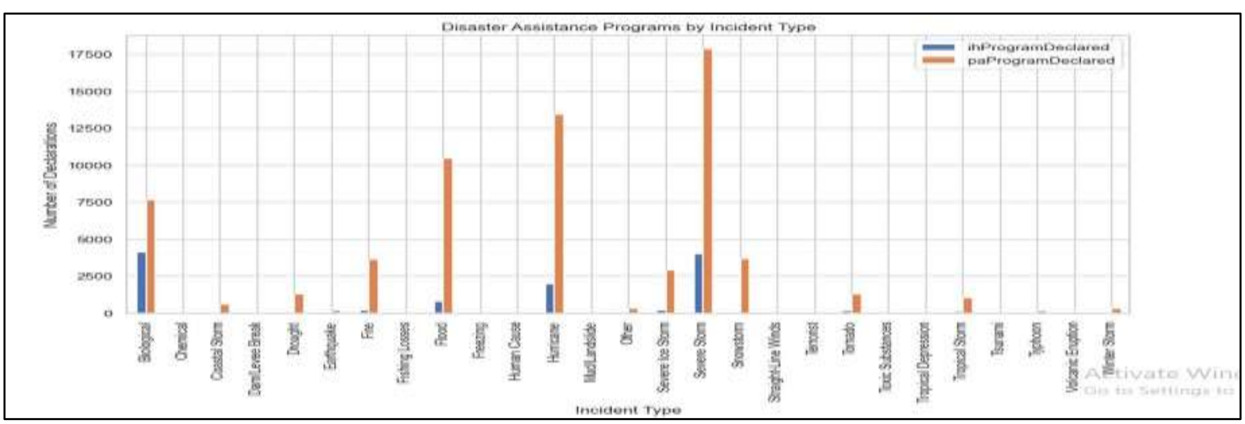


Interpretation

The heatmap represents disaster frequency across states and incident types using color intensity. Darker colors indicate higher disaster occurrence.
This visualization helps identify disaster hotspots and regional concentration patterns.

5.5 Disaster Assistance Programs by Incident Types

This bar chart illustrates the relationship between disaster types and assistance programs by aggregating ihProgramDeclared and paProgramDeclared for each incident type.

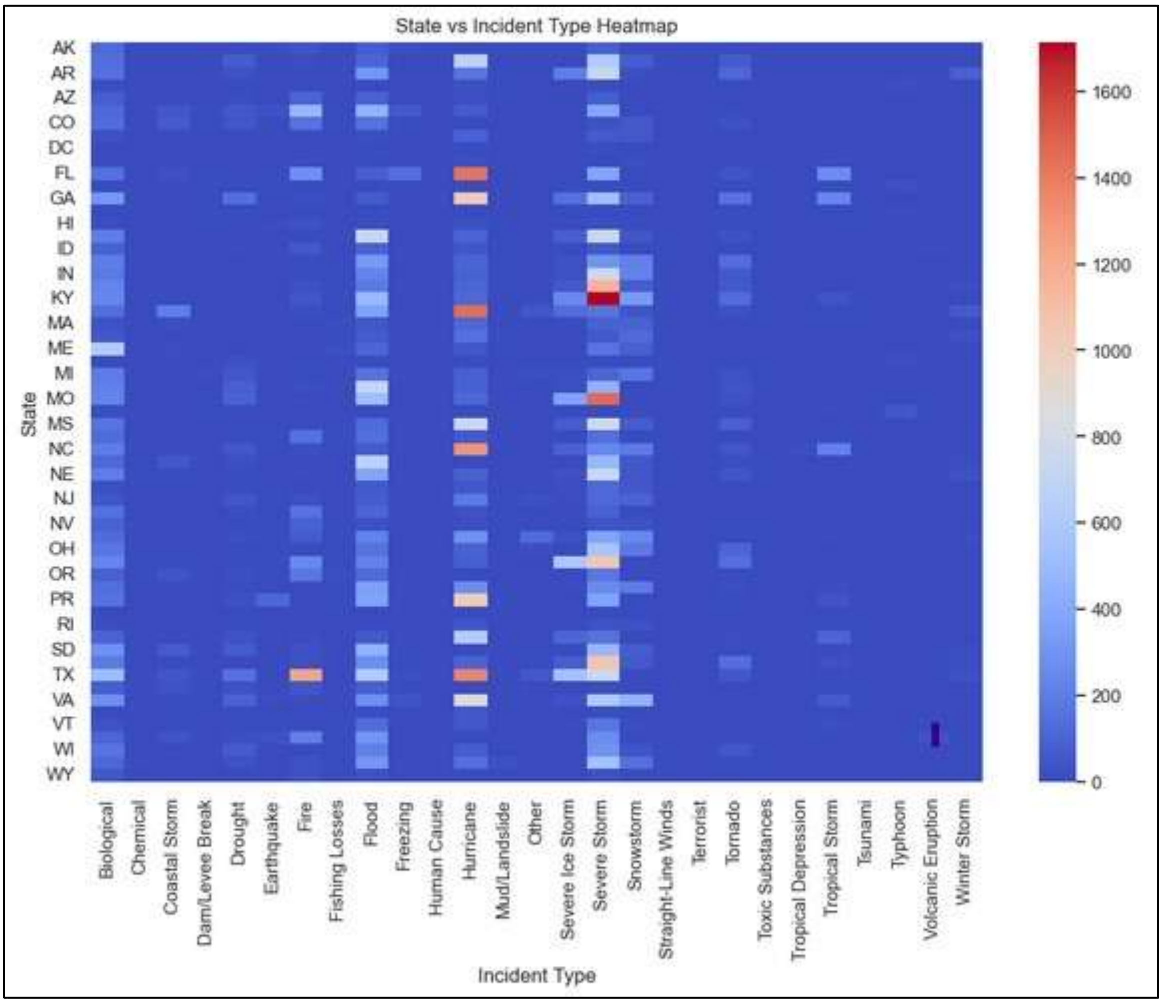


Interpretation

This bar chart shows how disaster types are associated with assistance programs.
Certain disaster types require more assistance compared to others.
It helps understand the impact and severity of different disasters.

5.6 State vs Incident Type Heatmap

This heatmap provides a detailed visualization of disaster intensity across states and incident types, highlighting variations using color gradients.

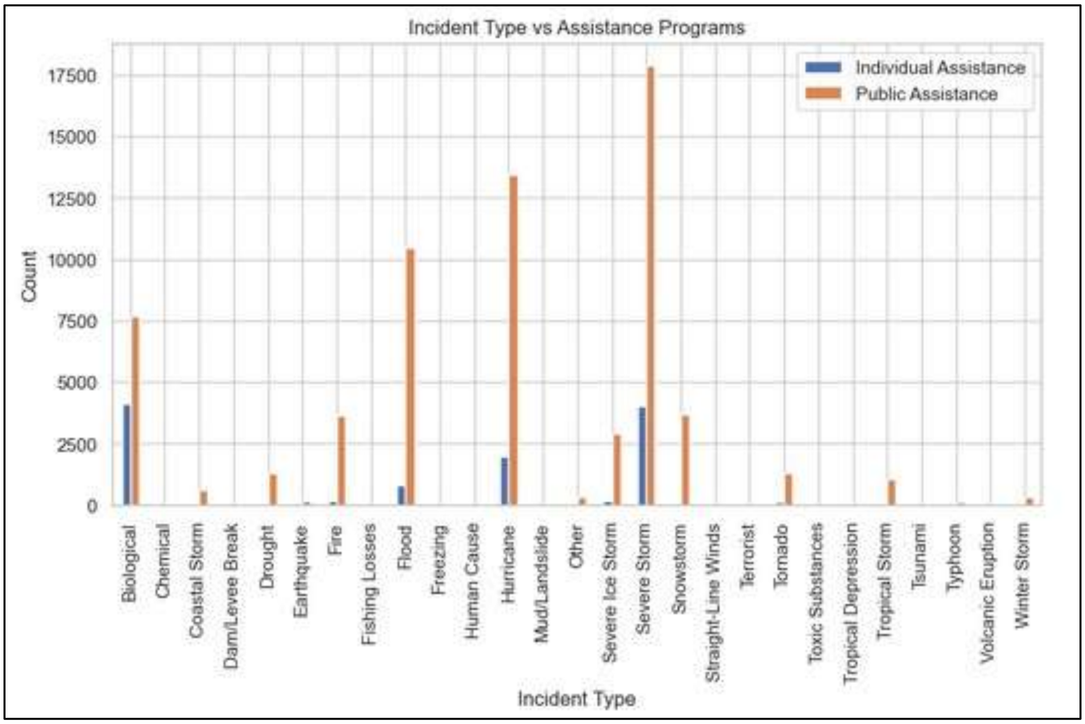


Interpretation

This heatmap provides deeper insights into disaster distribution across states and incident types.
It highlights intensity and concentration of disasters.
This visualization helps identify dominant disaster patterns at the state level.

5.7 Incident Type vs Assistance Programs

This grouped bar chart compares different disaster types with individual and public assistance programs to show how assistance varies by incident type.



Interpretation

This visualization compares incident types with individual and public assistance programs.
It shows how different disaster types influence assistance requirements.
Some disasters require more public assistance, while others impact individuals more directly.

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6. Tools and Libraries Used

The following tools and libraries were used for the analysis.

Data Processing

• Pandas – Used for data loading, cleaning, and aggregation
• NumPy – Used for numerical operations

Visualization

• Matplotlib – Used for basic charts
• Seaborn – Used for heatmaps and statistical visualizations
• Plotly – Used for interactive visualizations

These tools enabled effective analysis and visualization of disaster data.

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7. Key Insights

The incident type analysis reveals several important patterns:

• Certain disaster types occur more frequently than others
• Disaster types vary significantly across states
• Specific states are more prone to particular disaster types
• Disaster assistance programs vary depending on disaster type
• Visualizations clearly highlight disaster composition and impact

These insights help in understanding disaster behavior and response requirements.

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8. Conclusion

In Milestone-04 successfully analyzed disaster incident types and their distribution across states and assistance programs.

Through data aggregation and visualizations, the analysis identified dominant disaster types, regional variations, and the relationship between disaster types and assistance programs.

The results show that disaster occurrences follow clear patterns based on type and location, and assistance programs play a significant role depending on disaster severity.

This milestone integrates insights from previous analysis and provides a comprehensive understanding of disaster trends and impacts.

9. Deployed Dashboard Link

The interactive dashboard for this project has been successfully deployed and can be accessed using the link below:

Live Dashboard: https://u-s-natural-disaster-dashboard.vercel.app